Initial Gompertz analysis of mortality rate doubling time in BXD family females on chow and high-fat diets

What was done, step by step. Exploratory analysis, 23 October 2026.

Code and outputs: github.com/robwwilliams/bxd_mortality_rates_2026, folders Gompertz/ and output/ Read with Figures 1 to 4; legends are in section 8.

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1. Question and answer

Does a high-fat diet (HF) change how fast mortality accelerates with age, measured as the Gompertz slope and expressed as the mortality rate doubling time (MRDT)? Or does it change the age-independent level of mortality (R0)? Or both?

Answer: HF does not change MRDT: 153 d on chow (CD) and 149 d on HF, with p = 0.55 for equal slopes. HF raises mortality **1.42-fold at every age** (95% CI 1.29–1.55). That is equivalent to aging 76 days sooner, and the observed difference in median age at death is 72 days (Figures 1 and 3).

2. Data

- **Source:** Suheeta Roy's curated file of females used for actuarial mapping (BXD_Mortality_for_Mapping_Arends.xlsx), unchanged. The columns used are Diet, Strain, Age at death and Age at set up.
- **Included:** all CD and HF females, 1,934 animals in 124 strains (Table 1). **Excluded:** the 172 HF + 1% NAM females. The file contains no males and no BXD × BXD F1s.
- **No censoring:** every animal in the file died.
- **Left truncation:** each animal counts as at risk only from its age at set up, which ranges from 26 to 877 d. Animals that entered late therefore do not make the early ages look safer than they were.

Table 1. Females analyzed

Diet	BXD	C57BL/6J and DBA/2J	B6D2F1 and D2B6F1	Total
Chow (CD)	929	88	27	1,044
High fat (HF)	818	47	25	890

3. Model and fitting

We write the Gompertz law in base 2, so its parameters read directly. The hazard at age t (days) is

$$h(t) = \mathrm{R0} \cdot 2^{t^* / \mathrm{MRDT}},$$

where R0 is the mortality rate extrapolated to age 0 (per day), and mortality doubles every MRDT days. This is the same law as $h(t) = a\, e^{bt}$ with a = R0 and b = ln 2 / MRDT. The fitting code works in that natural-log form, which changes nothing in the results. For an animal that enters at age e and dies at age t, the log-likelihood is

$$\ln L = \ln h(t) - [H(t) - H(e)], \quad H(t) = (a/b)(e^{bt} - 1).$$

- **Fitting:** maximum likelihood with R's optim (BFGS) and analytic gradients. The parameters are ln a and ln b, so both stay positive.
- **Intervals:** 95% Wald intervals for MRDT come from the Hessian on the ln b scale.
- **Gompertz–Makeham:** this variant adds a constant hazard, $h(t) = a\, e^{bt} + c$.
- **Check on simulated data** (test_gompertz.R): 200 simulated studies of 1,000 animals each, with entry ages of 30–300 d and a true MRDT of 120 d.
  - The mean estimate was 119.7 d, and 93% of the 95% intervals contained 120 d. When the slopes were in fact equal, the equal-slope test rejected at p < 0.05 in 3.5% of runs.

4. Comparing the diets

- **Equal slope (equal MRDT):** a likelihood-ratio test (LRT) of separate fits (*a* and *b* for each diet) against a model with a separate *a* for each diet and one shared *b* (1 df).
- **Equal R0, given a shared slope:** an LRT of the shared-slope model against a single curve for both diets (1 df). In the shared-slope model, the ratio of the two *a* values is the HF/CD hazard ratio; its Wald CI comes from the Hessian.
- **One curve for both diets:** an LRT of separate fits against a single curve (2 df).
- **Cox models** (R survival package, left-truncated): the pooled hazard ratio, a Schoenfeld-residual test of proportional hazards (cox.zph), and a Cox model stratified by strain.
- **Bootstrap:** 2,000 resamples of animals, with replacement, within each diet (seed 20261023). Each resample is refitted. Percentile 95% intervals are taken for each MRDT and for the HF − CD difference.
- **Predicted lifespan shift:** with a shared slope, multiplying the hazard by the hazard ratio (HR) is the same as being $\log_2$(HR) doublings older, or MRDT × $\log_2$(HR) days.
- **Actuarial survivorships** (Figure 2b, after Arends et al. 2026, Fig. 1b): for each T-age from 200 to 1,100 d in 15-day steps, the mice alive and under observation at that age (set up by T, died at T or later). For each diet we take the mean age at death minus the mean for both diets, ± s.e.m., and the same quantity predicted by the Gompertz fits.

5. Sensitivity analyses (rows of Figure 3)

- **Survived to 200 d:** only animals that died at 200 d or later, each at risk from the later of its set-up age and 200 d. This removes 26 CD and 25 HF early deaths.
- **Survived to 365 d:** the same, at 365 d.
- **BXD strains only:** the parents and their F1s removed.
- **Gompertz–Makeham:** adds the constant hazard *c*.
- **Binned regression** (de Magalhães et al. 2018):
  - Deaths per mouse-day at risk in 30-day intervals, counting each mouse from its set-up age.
  - ln(rate) is regressed on the interval midpoint, weighted by deaths, using intervals from 200 d onward that contain at least one death.

6. Results

Table 2. MRDT in days (95% interval) by diet, and the test of equal slopes

Analysis	CD	HF	Equal slope, p
Primary (Wald)	153 (144–163)	149 (140–159)	0.55
Primary (bootstrap)	153 (143–164)	149 (132–165)	HF − CD −4 d (−24 to +15)
Survived to 200 d	152 (142–163)	152 (142–163)	0.99
Survived to 365 d	155 (142–168)	177 (160–196)	0.035
BXD strains only	156 (146–167)	149 (139–159)	0.31
Gompertz–Makeham	153 (144–163)	149 (140–159)	–
Binned regression from 200 d	152 (143–162)	144 (127–167)	–

- **Shared slope:** MRDT 151 d. HF/CD hazard ratio **1.42** (1.29–1.55), or **0.50 doublings**. The LRT for equal R0 gives p = 7×10^{-14}.
- **Cox:** HR 1.44 (1.32–1.58), with no departure from proportional hazards (p = 0.85). **Stratified by strain:** HR 1.78 (1.61–1.98).
- **Lifespan shift:** the predicted shift is 76 d; the observed difference in median age at death is 683 − 611 = 72 d.
- **Makeham term:** *c* is about 10^{-8} per day on both diets, which is negligible. The Gompertz–Makeham fits equal the Gompertz fits.

- **The R0–MRDT ridge:** R0 and MRDT estimates correlate at 0.95 (CD) and 0.97 (HF) across bootstrap fits (Figure 4).
- **Survivorship gap (Figure 2b):** HF survivors die 68 d younger than CD survivors at T200 and 65 d younger at T395. The gap narrows to 52 d at T605, 30 d at T710 and 6 d at T800, then reverses after about T860, with only 56 HF mice left at T860 and 24 at T935. The Gompertz fits predict a gap of 58 d at T200 that narrows only to 26 d at T800 and never reverses.

7. Interpretation and limits

- **R0, not slope.** The diets differ in R0, not in slope. This holds in five of the six analyses, including with all deaths before 200 d removed.
- **One analysis disagrees.** Among mice that survived to 365 d, HF has a shallower slope (MRDT 177 against 155 d, p = 0.035; proportional-hazards test p = 0.07). That is one nominal result among four uncorrected tests.
  - Two explanations are possible: population heterogeneity (a higher hazard removes frail mice sooner and flattens the late curve), or the R0–MRDT ridge.
- **Figure 2b shows where it happens.** The diet gap closes between about 650 and 850 d, much faster than a constant 1.42-fold hazard predicts, and reverses afterwards. Selective loss of frail HF mice would produce this pattern, and it is the same signal as the shallower HF slope after 365 d.
- **This is the MRDT of a mixture of 124 genotypes.** Pooling flattens the population curve and dilutes the diet effect: the HR is 1.44 pooled against 1.78 within strains. Roy et al. 2021 report 2.0.
- **Not yet adjusted** for vivarium (vivarium 3: 440 CD, 232 HF) or cohort. Animals still alive when the file was made are absent, which affects only the latest cohorts.
- **Next steps:** per-strain or per-type *a* and *b* with shrinkage; a gamma-Gompertz (frailty) model; vivarium as a covariate; a rerun on the approved data set.

8. Figure legends

Figure 1. Mortality rises in parallel on chow and high-fat diets. Mortality rate against age for chow-fed (CD, brown, n = 1,044) and high-fat-fed (HF, blue, n = 890) BXD family females. The y axis is deaths per 1,000 mouse-days on a log2 scale, so each gridline marks one doubling. Points are occurrence/exposure rates in 30-day age intervals: deaths in the interval divided by mouse-days at risk, each mouse counted from its age at set up. Intervals are shown where at least 20 mice were at risk; point diameter grows with the square root of the number of deaths. Lines are Gompertz fits by maximum likelihood, left-truncated at set-up age, drawn from 100 to 1,250 d. The key gives each diet's MRDT with its 95% Wald interval. The slopes do not differ (likelihood-ratio test, p = 0.55). With a shared slope (MRDT 151 d), HF mortality is 1.42 times CD mortality at every age (95% CI 1.29–1.55), which is 0.50 doublings higher.

Figure 2. Survival from 200 days (a) and the diet gap by T-age (b). (a) Kaplan–Meier survival (solid lines) of mice alive at 200 d (CD n = 1,018, HF n = 865), each at risk from the later of its set-up age and 200 d. Dashed lines show survival predicted by the primary Gompertz fits (Figure 1), conditional on survival to 200 d. Labels give median age at death among these mice. (b) The same data as actuarial survivorships, in the style of Arends et al. 2026, Fig. 1b. Each survivorship is the set of mice alive and under observation at a given T-age, that is, set up by that age and dying at or after it, in 15-day steps from 200 d. Lines show each diet's mean age at death minus the mean for both diets; shading shows ±1 s.e.m. Lines are drawn while both diets keep at least 20 mice (to 935 d); numbers along the top give mice alive (CD / HF). HF survivors die 65 to 69 d younger than CD survivors through 400 d. The gap closes between about 650 and 850 d, faster than the Gompertz fits predict (section 6), and reverses after about 860 d, when few HF mice remain.

Figure 3. MRDT is robust to the choice of analysis. Mortality rate doubling time (days) for CD (brown) and HF (blue) under the primary analysis and five alternatives (rows; definitions in section 5). Points are estimates and bars are 95% intervals. The intervals are Wald intervals on the log scale of the slope, except in two rows: the bootstrap row uses the 2.5th and 97.5th percentiles of 2,000 resamples, and the binned-regression row uses the regression standard error of the slope. At right is the likelihood-ratio p value for equal slopes in the four analyses where it was computed. Only the analysis restricted to mice surviving to 365 d gives a nominally different slope (p = 0.035).

Figure 4. R0 and MRDT estimates trade off along a ridge. Each small point is one of 2,000 bootstrap fits per diet (mice resampled with replacement within diet), plotting R0 against MRDT. R0 is deaths per 1,000 mouse-days at age 0, on a log2 scale. Large points are the estimates from the full data. Within each diet the estimates lie on a narrow diagonal ridge: R0 and MRDT estimates correlate at 0.95 (CD) and 0.97 (HF), the Strehler–Mildvan degeneracy of the Gompertz fit. The two ridges overlap in MRDT but are separated along R0, showing that the diets differ in age-independent mortality, not in the rate of aging. Because of the ridge, slopes should be compared with the joint likelihood-ratio test rather than by comparing separate intervals.

9. Reproducing

From the repository root:

```
Rscript Gompertz/test_gompertz.R; Rscript Gompertz/MRDT_Diet.R; Rscript
Gompertz/Figures_MRDT_Diet.R; python3 tools/svg2png.py output/*.svg
```

R 4.3 with readxl, survival and svglite. The SVG files keep each label as one editable text element in Arial. The PNG previews are rendered with CairoSVG at 150 dpi.